TRIPATHI STUDY ZONE
JEE & NEET
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CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 5

(With Solutions)

Duration : 3 Hours **Max. Mark : 720**

Instructions

- The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- The question paper contains three parts of Physics, Chemistry and Biology respectively.
- Part A contains 45 Questions from Physics section. Part B contains 45 Questions from Chemistry section. Part C contains 90 Questions from Biology.

PART A PHYSICS

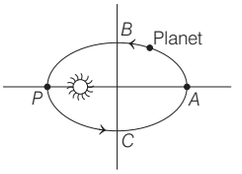
1. The forbidden energy gap between valence and conduction bands of Ge and Si are respectively,
(a) 5 eV, 1.1 eV (b) 0.78 eV, 1.1 eV
(c) 1.1 eV, 0.38 eV (d) 0.78 eV, 0.78 eV

2. A rectangular frame of copper wire enclosing a surface area A is covered with a soap film. If the area of the frame of copper wire is reduced 25%, the energy of the soap film will be changed by
(a) 25% (b) 50%
(c) 75% (d) 35%

3. A ray of light falls on a transparent glass slab of refractive index 1.73. If the reflected ray and the refracted ray are mutually perpendicular, the angle of incidence is
(a) 30° (b) 48°
(c) 60° (d) 75.2°

4. Suppose the speed of planet at the perihelion P as in the figure be v_P and the sun-planet distance SP be r_P.

Then, (r_P, v_P) are related to the corresponding quantities at the aphelion (r_A, v_A) as

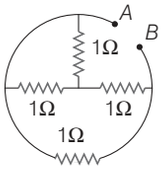


(a) $r_P > r_A$, $v_P > v_A$ (b) $r_P < r_A$, $v_P > v_A$
(c) $r_A > r_P$, $v_A > v_P$ (d) $r_A < v_P$, $v_A = v_P$

5. The path difference for constructive interference in terms of wavelength λ is
(a) $n\lambda$ (b) $(2n+1)\lambda$
(c) $(2n-1)\lambda$ (d) $(n+1)\frac{\lambda}{2}$

6. A force F acting on a body depends on its displacement s as $F \propto s^{1/3}$. The power delivered by F will depend on displacement as
(a) $s^{2/3}$ (b) $s^{1/2}$ (c) s (d) s^0

7. The equivalent resistance between A and B as shown in the figure,



(a) 0.2Ω (b) 0.4Ω
(c) 0.6Ω (d) None of these

8. Three balls of masses 4kg, 3 kg and 5 kg are arranged at the corners of an equilateral triangle ΔABC of side 2 m. The moment of inertia of the system about an axis through the centroid and perpendicular to the plane of the triangle is
(a) 16 kg-m^2 (b) $4\sqrt{3}$ kg-m^2 (c) 4 kg-m^2 (d) $16\sqrt{3}$ kg-m^2

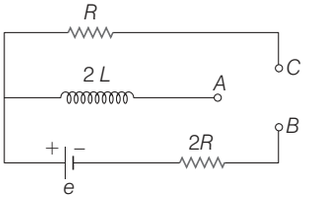
9. A block of mass 10 kg is moving horizontally with a speed of 1.5 ms^{-1} on a smooth plane. If a constant vertical force of 10 N acts on it, then the displacement of the block from the point of application of the force at the end of 4s is
(a) 5 m (b) 20 m (c) 12 m (d) 10 m

10. A copper wire has a mass 0.3 ± 0.003 g, radius 0.5 ± 0.005 mm and length 6 ± 0.06 cm. The maximum percentage error in the measurement of its density is
(a) 2% (b) 3% (c) 6% (d) 4%

11. A square loop of side 20 cm having resistance 50 Ω is placed in an uniform magnetic field of 2.5 T to the plane of loop. If we attempt to pull it out of the field with a constant velocity 25 m/s, then the power requirement is
(a) 2.75 W (b) 3.75 W (c) 3.125 W (d) 5.125 W

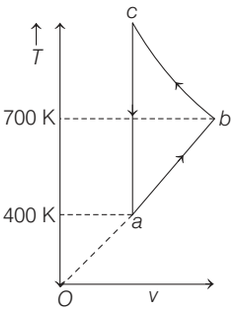
12. A brass rod 1 m long is firmly clamped in the middle and one end is stroked by a resined cloth. What is the pitch of the note will be heard? (Young's modulus for brass = 10^2 dyne cm^{-2} and density = 9 g cm^{-3})
(a) 1652.5 Hz (b) 1600 Hz (c) 1666.67 Hz (d) 1554.48 Hz

13. In the circuit shown below, A is joined to B for a long and then, A is joined to C. The total heat produced in R is



(a) $\frac{Le^2}{8R^2}$ (b) $\frac{Le^2}{2R^2}$ (c) $\frac{Le^2}{4R^2}$ (d) $\frac{Le^2}{R^2}$

14. A cyclic process $abca$ performed on 3 moles of an ideal gas. A total of 1500 cal of heat is withdrawn from the gas sample in the process. The work done by the gas during the path bc is



(a) –3000 cal (b) + 3000 cal
(c) – 1500 cal (d) – 3300 cal

15. The value of refractive index of a medium of polarising angle of 76º is [given, tan 76º = 4]
(a) 2 (b) 4
(c) 1.5 (d) None of these

16. A body hanging from a spring balance stretches it by 1 cm at the earth's surface. The same body stretches the spring at a place 3200 km below the surface of earth by a value of
(take, radius of earth = 6400 km)
(a) 2 cm (b) 0.5 cm
(c) 1 cm (d) 6 cm

17. The time period of the electron orbiting in second orbit of hydrogen atom is 4×10^{-16}s. Hence, the time period in the third orbit will be
(a) 2.7×10^{-16} s (b) 1.35×10^{-16} s
(c) 2.7×10^{-15} s (d) 1.35×10^{-15} s

18. Treating the bulb as a point source of electromagnetic radiation, i.e. about 5% efficient in converting electrical energy to visible light. The maximum magnitudes of the electric field of the light that is incident on a page, because of the visible light coming from a desk lamp is
(a) 40 V/m (b) 30 V/m
(c) 45 V/m (d) 25 V/m

19. A metal rod of length 2.5 m and negligible mass has two masses tied to its ends. If they have masses 0.5 kg and 1.5 kg, respectively, then find the location of the point on this rod, where the rotational energy is minimum, when rod is rotated about that point.
(a) 1.875 m from 1.5 kg
(b) 1.525 m from 1.5 kg
(c) 1.525 m from 0.5 kg
(d) 1.875 m from 0.5 kg

20. Monochromatic light of wavelength 660 nm is produced by a helium neon-laser. The power emitted is 9 mW. The number of photons emitted per second is

(a) 4.5×10^{16} /s (b) 3×10^{16} /s

(c) 3.6×10^{16} /s (d) 2×10^{16} /s

21. A car accelerates from rest at a constant rate α for sometime after which it deaccelerates at constant rate β and finally comes to rest. If total time elapsed is T_1, then maximum velocity attained by car is

(a) $\frac{\alpha+\beta}{\alpha\beta}T$ (b) $\frac{\alpha\beta}{\alpha+\beta}T$

(c) $\frac{\alpha}{\alpha+\beta}T$ (d) $\frac{\beta}{\alpha+\beta}T$

22. The Young's modulus of the material of a wire is 5×10^{10} N/m^2. If the elongation strain is 2%, then the energy stored in the wire per unit volume (in Jm^{-3}) is

(a) 10^7 (b) 2×10^7

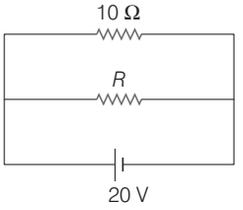
(c) 0.5×10^7 (d) 10^8

23. Which of the following is not a path function?

(a) ΔQ (b) $\Delta Q - \Delta W$

(c) ΔW (d) $nRT\log_e\left(\frac{V_2}{V_1}\right)$

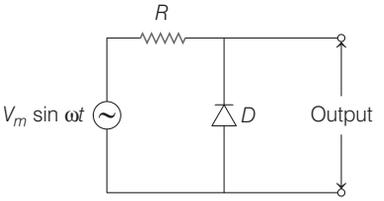
24. The power dissipated in the circuit shown below is 50 W, then the value of R is



(a) 20Ω (b) 30Ω

(c) 40Ω (d) 50Ω

25. The output of the given circuit in the figure



(a) would be zero at all times

(b) would be like that of a full-wave rectifier

(c) would be like a half- wave rectifier with positive cycle in output

(d) would be like a half- wave rectifier with negative cycle in output

26. Infinite number of masses, each of 1 kg are placed along the X-axis at $x = \pm1$m, ±2m, $\pm$ 4m, $\pm$ 8m, ±16m, ... The magnitude of the resultant gravitational potential in terms of gravitational constant G at the origin ($x = 0$) is

(a) G (b) $2G$ (c) $3G$ (d) $4G$

27. A raw egg and a hard boiled egg are made to spin on a table with the same angular speed about the same axis. The ratio of time taken by the two eggs to stop is

(a) equal to 1 (b) less than 1

(c) greater than 1 (d) zero

28. When temperature of radioactive substance increases, then its half-life

(a) increases (b) decreases

(c) remains same (d) None of these

29. A ray of light strikes a materials slab at an angle of incidence 30°. If the reflected and refracted rays are perpendicular to each other, the refractive index of the materials is

(a) $\sqrt{3}$ (b) $\sqrt{2}$ (c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{\sqrt{3}}$

30. Charge q_2 of mass m revolves around a stationary charge q_1 in a circular orbit of radius r. The orbital periodic time of q_2 would be

(a) $\left[\frac{4\pi^2mr^2}{q_1q_2}\right]^{\frac{1}{2}}$ (b) $\left[\frac{kq_1q_2}{4\pi^2mr^2}\right]^{\frac{1}{2}}$ (c) $\left[\frac{4\pi^3mr^2}{kq_1q_2}\right]^{\frac{1}{2}}$ (d) $\left[\frac{4\pi^2mr^4}{q_1q_2}\right]^{-\frac{1}{2}}$

31. The width of one slit is double the another slit in Young's double slit experiment, then the ratio of maximum intensities to minimum intensities in the interference pattern is

(a) 9 : 1 (b) 1 : 9 (c) 3 : 1 (d) 1 : 3

32. A policeman blows a whistle with a frequency of 500 Hz. A car approaches him with a velocity of 15ms^{-1}. The change in frequency as heard by the driver of the car as he passes the policeman is (speed of sound in air = 300 ms^{-1})

(a) 25 Hz (b) 50 Hz (c) 75 Hz (d) 45 Hz

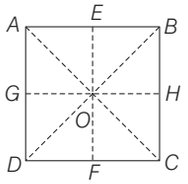
33. A pure Si wafer is doped with Al. The number density of acceptor atoms is approximately 10^{22}m^{-3}, if density of the hole in an intrinsic semiconductor is approximately 10^{20}m^{-3}. The number density of electron in the wafer is

(a) 10^4m^{-3} (b) 10^{12}m^{-3} (c) 10^{18}m^{-3} (d) 10^{24}m^{-3}

34. 20 g of water at 50°C and 8 g of ice at –20° C are mixed together in a calorimetry. What is the final temperature of the mixture? (Given, specific heat of ice = 0.5 cal g^{-1} °C^{-1} and latent heat of fusion of ice = 80 cal g^{-1})

(a) –5 °C (b) 5°C (c) 20°C (d) 28°C

35. For the given an uniform square lamina $ABCD$, whose centre is O

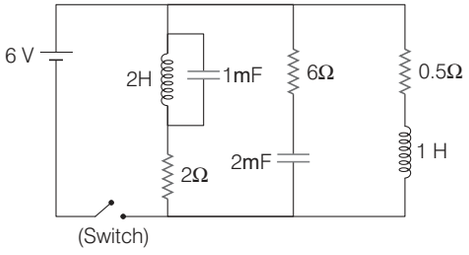


(a) $\sqrt{2}I_{AC} = I_{EF}$ (b) $I_{AD} = 3I_{EF}$ (c) $I_{AC} = \sqrt{2}I_{EF}$ (d) $I_{AC} = I_{EF}$

36. An uncharged capacitor with a solid dielectric is connected to a similar air capacitor charged to a potential of 50 V. If the common potential after sharing the charges becomes 2V, then the dielectric constant of dielectric must be

(a) 16 (b) 24 (c) 36 (d) 20

37. In the following circuit, current through 2 Ω resistor just after switch closed is



(a) 3 A (b) 2.25 A (c) 0.71 A (d) 0A

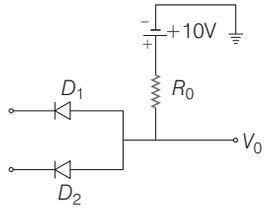
38. Match the corresponding entries of Column I with Column II and choose the correct option from the codes given below

	Column I		Column II
A.	First law of thermodynamics	P.	Path dependent
B.	ΔW	Q.	$\Delta U = 0$
C.	Isothermal process	R.	$p\Delta V$
D.	ΔQ	S.	Conservation of energy

Codes

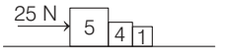
	A	B	C	D		A	B	C	D
(a)	S	Q	R	P	(b)	S	P	R	Q
(c)	S	R	Q	P	(d)	S	Q	P	R

39. The name of the following gate that the circuit shown in the figure



(a) AND gate (b) OR gate (c) NOT gate (d) NOR gate

40. Three blocks of mass 5 kg, 4 kg and 1 kg, respectively are in contact on a frictionless table as shown in the figure. If a force of 25 N is applied on the 5 kg block, the contact force between 5 kg and 4 kg block will be



(a) 12.5 N (b) 12 N
(c) 2.5 N (d) 10 N

41. A bar magnet is oscillating in the earth's magnetic field with a time period T. What happens to its time period and motion, if its mass is quadrupled?

(a) Motion remains SHM and time period remains nearly constant
(b) Motion remains SHM with time period = $2T$
(c) Motion remains SHM with time period = $4T$
(d) Motion remains SHM and time period = $T/2$

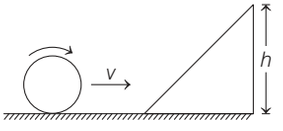
42. The length of a simple pendulum is about 100 cm known to an accuracy of 1 mm. Its period of oscillation is 2s, determined by measuring the time for 100 oscillations using a clock of 0.1 s resolution. What is the accuracy in the determined value of g ?

(a) 0.5% (b) 0.25%
(c) 0.2% (d) 1.0%

43. The work function of metal is in the range of 2eV to 5eV. Then which of the following wavelength of light cannot be used for photoelectric effect. (take, $h = 4 \times 10^{-15}$ eV, $c = 3 \times 10^{8}$ m/s)

(a) 510 nm
(b) 650 nm
(c) 400 nm
(d) 570 nm

44. A solid ball is rolling on a frictionless horizontal surface with a translational velocity v ms^{-1} as shown in the figure. If it has to climb the inclined surface, then its velocity should be



(a) $\geq \sqrt{2gh}$ (b) $\sqrt{2gh}$
(c) $\leq \sqrt{10gh/7}$ (d) $\geq \sqrt{10gh/7}$

45. The velocity of a particle describing SHM is 16 cms^{-1} at a distance of 8 cm from mean position and 8 cms^{-1} at a distance of 12 cm from mean position. The amplitude of motion

(a) 15.5 cm (b) 13 cm
(c) 14.3 cm (d) 13.06 cm

PART B CHEMISTRY

46. The general molecular formula of the products obtained on heating lanthanoids (Ln) with sulphur is

(a) Ln_2S_3 (b) Ln_3S_2
(c) LnS (d) LnS_3

47. The half-life time of a first order reaction is 1.26×10^{14} s, then rate constant of this reaction will be

(a) $5.5\times 10^{-10}\ s^{-1}$ (b) $5.5\times 10^{-15}\ s^{-1}$
(c) $6.6\times 10^{-14}\ s^{-1}$ (d) None of these

48. Equivalent conductance of saturated $MgSO_4$ is $380\ \Omega^{-1}\ cm^2\ equiv^{-1}$ and specific conductance is $5\times 10^{-4}\ \Omega^{-1}\ cm^{-1}$. Thus, K_{sp} of $MgSO_4$ is

(a) $8.38\times 10^{-10}\ M^2$ (b) $43.56\times 10^{-8}\ M^2$
(c) $59.10\times 10^{-4}\ M^2$ (d) $23.25\times 10^{-8}\ M^2$

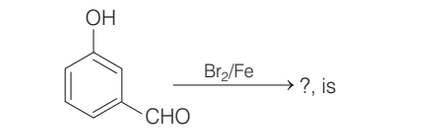
49. Which of the following element does not belong to $4d$ series of transition elements?

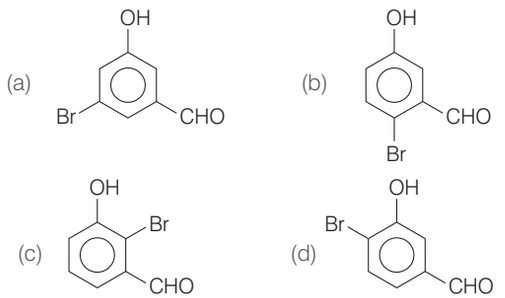
(a) Molybednum (b) Titanium
(c) Technetium (d) Zirconium

50. Which of the following pair of solution is expected to show isotonic at same temperature?

(a) 0.2 M urea and 0.2 M KCl
(b) 0.1 M urea and 0.2 M $CaCl_2$
(c) 0.1 M KCl and 0.1 M Na_2SO_4
(d) 0.1 M $Ca(NO_3)_2$ and 0.1 M Na_2SO_4

51. The product of the following reaction,





52. Which of the following is used as an antifreeze in cars for cooling the engine?

(a) 10% ethanol solution in water
(b) 35% solution of ethylene glycol
(c) 45% solution of ethylene glycol
(d) 20% ethanol solution in water

53. Which of the following coordination compound would exhibit optical isomerism?

(a) Pentaammine nitro cobalt (III) iodide
(b) Diammine dichloroplatinum (II)
(c) *trans* dicyano *bis* (ethylene diamine) chromium (III) chloride
(d) *tris*-(ethylene diamine) cobalt (III) bromide

54. Consider the reaction, $2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$. If the concentration of NO_2 increases by 8.5×10^{-3} M in 100 s, then the rate of reaction will be

(a) $1.3\times 10^{-5}\ Ms^{-1}$ (b) $2.8\times 10^{-5}\ Ms^{-1}$
(c) $7.6\times 10^{-4}\ Ms^{-1}$ (d) $2.13\times 10^{-5}\ Ms^{-1}$

55. CFCs, HCFCs and halons destroy the earth's protective ozone layer, which shields the earth from harmful ultraviolet rays generated from the sun. Why CFC are widely used in air conditioners?

(a) Because CFC are highly reactive
(b) It is highly inflammable
(c) It is non-reactive
(d) All of the above are true

56. Which of the following statements is incorrect for cathode rays?

(a) The cathode rays start from cathode and move towards the anode
(b) In the absence of electrical or magnetic field, these rays travel in straight lines
(c) The characteristics of cathode rays depend on the material of electrodes and the nature of the gas present in the cathode rays tube
(d) Cathode rays consist of negatively charged particles

57. The bond between carbon atom (1) and carbon atom (2) in compound $N\equiv \overset{1}{C}-\overset{2}{C}H=CH_2$ involves the hybrid orbitals

(a) sp^2 and sp^2 (b) sp^3 and sp (c) sp and sp^2 (d) sp and sp

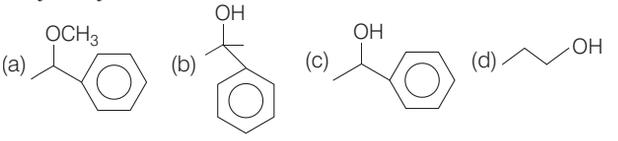
58. IP_1 and IP_2 of Mg are 178 and 348 kcal mol^{-1}. The energy required for the reaction, $Mg \longrightarrow Mg^{2+} + 2e^-$ is

(a) + 170 kcal (b) + 526 kcal (c) – 170 kcal (d) – 526 kcal

59. Which of the following has strongest hydrogen bonding?

(a) O—H ...N (b) F—H ...F (c) O—H ...O (d) O—H ...F

60. The product formed when acetone is treated with phenyl magnesium bromide followed by acidic hydrolysis is



61. Consider the following statement about chelating ligands:

I. They have more than one donor atom.

II. They are non-linear and they must form bonds with reasonable angles at the metal atom.

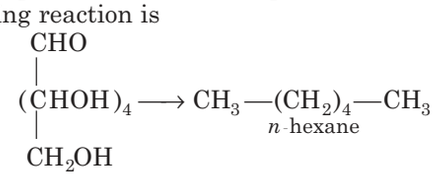
III. Ethylenediamine is a bidentate chelating ligand.

The correct statement is/are

(a) Only I (b) Both I and II

(c) Both II and III (d) I, II and III

62. The reagent and condition required for the following reaction is



(a) HI, Δ (b) HBr, Δ

(c) HF, Δ (d) HCl, Δ

63. In order to prepare a 1° amine from an alkyl halide with simultaneous addition of one $>CH_2$ group in the carbon chain, the reagent used as a source of nitrogen is

(a) potassium phthalimide, $C_6H_4(CO)_2N^-K^+$

(b) potassium cyanide, KCN

(c) sodium azide, NaN_3

(d) sodium amide, $NaNH_2$

64. If 8 g of NaOH is dissolved in water to make 1 L of solution, then the pH of solution is

(a) 13.70 (b) 10.28

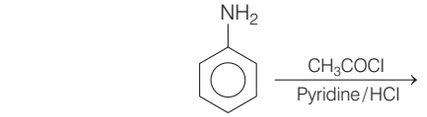
(c) 8.42 (d) 12.89

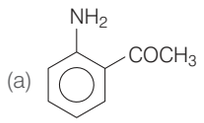
65. The number of chain isomers of C_7H_{16} are

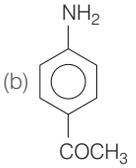
(a) 6 (b) 4

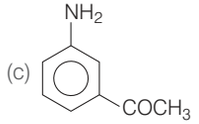
(c) 9 (d) 7

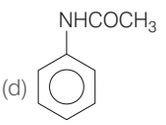
66. Product of the following reaction is



(a) 

(b) 

(c) 

(d) 

67. The pK_a of a weak acid H*A* is 4.92. The pK_b of a weak base *B*OH is 4.81. The pH of an aqueous solution of the corresponding salt, *BA*, will be

(a) 7.05 (b) 8.29

(c) 9.58 (d) 4.79

68. Consider the following compounds :

I. *cis*-$[Co(NH_3)_4Cl_2]^+$ II. *trans*-$[Co(en)_2Cl_2]^+$

III. *cis*-$[Co(en)_2Cl_2]^+$ IV. $[Co(en)_3]^{3+}$

Which of the given compound show optical isomerism?

(a) Both I and II (b) Both II and III

(c) Both III and IV (d) I, III and IV

69. Which of the following order is correct for the acidic nature of the compounds?

(a) $CH_3OH < CH_3SH < CH_3NH_2$

(b) $CH_3NH_2 < CH_3OH < CH_3SH$

(c) $CH_3SH < CH_3OH < CH_3NH_2$

(d) $CH_3SH < CH_3NH_2 < CH_3OH$

70. Within each transition series, the oxidation state

(a) first decreases till the middle of period and then increases

(b) first increases till the middle of the period and then decreases

(c) decreases regularly on moving from left to right

(d) None of the above

71. Which of the following statement(s) is incorrect for oxides of chlorine?

(a) ClO_3 forms dimer due to unpaired electron

(b) Cl_2O_6 is the anhydride of $HClO_3$

(c) ClO_3 is paramagnetic and exists in liquid state

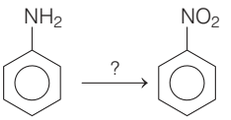
(d) Cl_2O_6 is diamagnetic in solid state

72. In the estimation of nitrogen by Kjeldahl's method, 2.8 g of an organic compound required 20 millimole of H_2SO_4 for the complete neutralisation of NH_3 gas evolved. The percentage of nitrogen in the sample is

(a) 20% (b) 10%

(c) 40% (d) 30%

73. Which of the following reagents will be used to complete below conversion?



(a) $KMnO_4$ (b) H_2SO_4

(c) P_2O_5 (d) CF_3COOOH

74. When conc. H_2SO_4 is heated with P_2O_5, the acid is converted to

(a) a mixture of sulphur dioxide and sulphur trioxide

(b) sulphur

(c) sulphur dioxide

(d) sulphur trioxide

75. A flux is chemical leaning agent, flowing agent or purifying agent. They may have more than one function at a time. They are used in both extractive metallurgy and metal joining. Which of the following is an example of acidic flux?
(a) SiO_2 (b) CaO
(c) MgO (d) None of these

76. The electronic configurations of Cr^{2+}, Mn^{2+}, Fe^{2+} and Ni^{2+} are $3d^4$, $3d^5$, $3d^6$ and $3d^8$ respectively. Which one of the following aqua complexes will exhibit the minimum paramagnetic behaviour?
(a) $[Mn(H_2O)_6]^{2+}$ (b) $[Fe(H_2O)_6]^{2+}$
(c) $[Ni(H_2O)_6]^{2+}$ (d) $[Cr(H_2O)_6]^{2+}$

77. Phosphorus trichloride PCl_3, undergoes hydrolysis to form an oxoacid. The formula of this oxoacid is
(a) H_3PO_2 (b) H_3PO_4 (c) HPO_3 (d) H_3PO_3

78. Arrhenius proposed a quantitative relationship between rate constant and temperature as
$k = Ae^{\frac{-E_a}{RT}}$ or $\ln k = \ln A - E_a / RT$.
What will be the slope of graph when $\ln k$ *versus* $1/T$ is plotted?
(a) $\frac{E_a}{R}$ (b) $-\frac{E_a}{R}$ (c) $\frac{1}{T}$ (d) $\ln A$

79. Low level of noradrenaline is the cause of depression. The drug used to cure this problem is
(a) antidepressants (b) hypnotics
(c) sedatives (d) psychedic

80. Anomalous property of lithium is due to (Which of the below is most correct choice?)
(a) smaller size of metal ion
(b) smaller charge density
(c) smaller size and higher charge density
(d) None of the above

81. The vapour density of a gas is 11.2. The volume occupied by 11.2 g of this gas at N.T.P is
(a) 1 L (b) 11.2 L (c) 22.4 L (d) 20 L

82. $\Delta E°$ of combustion of *iso*-butylene is $-x$ kJ mol^{-1}. The value of $\Delta H°$ is
(a) $= \Delta E°$ (b) $> \Delta E°$ (c) = zero (d) $< \Delta E°$

83. The repeating unit present in asbestos is
(a) $[Si_4O_{11}]^{6-}$ (b) $[SiO_4]^{4-}$ (c) $[Si_2O_7]^{2-}$ (d) $[Si_3O_9]^{3-}$

84. Consider the following reaction, $H_2O(s) \rightleftharpoons H_2O(l)$
The water and ice is at equilibrium at 273 K and enthalpy change of process is 6 kJ mol^{-1}. The entropy change of conversion of ice into liquid water is
(a) 21.98 JK^{-1} mol^{-1} (b) – 21.98 JK^{-1} mol^{-1}
(c) – 21.98 JK^{-1} (d) + 21.98 kJ K^{-1}

85. Match Column I with Column II, identify correct answer by using the codes given below.

	Column I		**Column II** (Entropy)
A.	Endothermic process	1.	$\Delta S = nF\left(\frac{\Delta E_{cell}}{\Delta T}\right)$
B.	Residual entropy	2.	$dS = 0$
C.	Entropy of cell reaction	3.	$\Delta S > 0$
D.	Cyclic process	4.	$S_R = k\ln W = R\ln W$

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	4	1	2
(d)	4	1	3	2

86. Which of the following arrangement is possible?

	n	*l*	*m*	*s*		*n*	*l*	*m*	*s*
(a)	5	2	2	$+\frac{1}{2}$	(b)	2	2	0	$-\frac{1}{2}$
(c)	3	–2	1	$+\frac{1}{2}$	(d)	0	0	1	$+\frac{1}{2}$

87. Which of the following statement(s) is/are correct regarding carbonyl group?
(a) The three groups attached to the carbonyl carbon lie in different planes
(b) The three groups attached to the carbonyl carbon lies in the same plane
(c) The carbonyl carbon is *sp*-hybridised
(d) The carbonyl carbon is sp^3-hybridised

88. Which of the following statement is correct regarding elements of group 18?
(a) Helium can diffuse through rubber, PVC and even glass
(b) Ar, Kr and Xe form clathrate
(c) He has the lowest melting point and boiling point
(d) All of the above are correct statements

89. Doping can be done with an impurity which is electron rich or electron deficient as compared to the intrinsic semiconductor, silicon or germanium. Doping of silicon with boron leads to
(a) *n*-type semiconductor
(b) *p*-type semiconductor
(c) metal
(d) insulator

90. Which of the following reactants are used to obtain methylphenyl ether?
(a) Phenolate ions and methyl iodide
(b) Methoxide ions and bromobenzene
(c) Methanol and phenol
(d) Bromobenzene and methylbromide

PART C BIOLOGY

91. The drug whose plant leaves contain atropine and is used by doctors to dilate pupil of eye is
(a) reserpine (b) aspirin
(c) codein (d) belladonna

92. Which of the following enzymes is the replicative polymerase in *E. coli* ?
(a) DNA polymerase IV (b) DNA polymerase II
(c) DNA polymerase III (d) DNA polymerase I

93. Transposons are sequences of
(a) polypeptide chain (b) RNA
(c) DNA (d) chromosomes

94. Hypothyroidism is characterised by all of the following except
(a) coarse dry skin
(b) Gull's disease
(c) increased demand for energy
(d) heat intolerance

95. *Penicillium* undergoes sexual reproduction by the
(a) formation of cleistothecia
(b) formation of conidiophores
(c) formation of buds
(d) fragmentation

96. Which one of the following diseases is an arthropod borne disease ?
(a) Syphilis (b) Leprosy
(c) Typhus fever (d) Cholera

97. In human eyes, retina has cones which are responsible for the coloured vision and are sensitive to entire visible spectrum. The three principal cones are
(a) red, blue and green (b) violet, indigo and blue
(c) red, orange and yellow (d) red, blue and yellow

98. Which among the following is not an advantage of GM crops ?
(a) GM plants are able to tolerate cold climate
(b) GM plants can withstand long periods of drought
(c) GM plants help to eliminate the use of chemical pesticides
(d) GM plants cause unintended harm to other organisms

99. Which of the following statements is incorrect ?
(a) JBS Haldane — Natural selection theory
(b) de Vries — Mutation theory
(c) Lamarck — Inheritance of acquired characters
(d) Louis Pasteur — Germ theory

100. Non-endocrine hormones are secreted by the cells and widely spread in the body. Which of the following does not represent such hormone?
(a) Insulin (b) Motilin
(c) Ghrelin (d) Calcitonin

101. Which of the following ideas was not a part of Charles Darwin's theory of 'origin of species by natural selection'?
(a) More offsprings are produced by organism than the number an environment can support
(b) Gene mutation is responsible for variation among individuals
(c) Individuals best adapted to environment can successfully survive and reproduce
(d) Individuals of an area tend to compete for scarce resources

102. Which of the following genetic disorders runs in the British royal family?
(a) Haemophilia (b) Phenylketonuria
(c) Thalassemia (d) Colour blindness

103. Which method is used to analyse inheritance pattern of traits in humans?
(a) Karyotyping (b) DNA fingerprinting
(c) Pedigree analysis (d) All of these

104. The occurrence of haemophilia is more likely in human males compared to females because
(a) females are heterogametic
(b) females can only be carriers
(c) the haemophilic allele is present only on X-chromosome
(d) the haemophilic allele is present only on Y-chromosome

105. Rh factor is named after
(a) mouse (b) human
(c) monkey (d) rabbit

106. Certain glands produce hormones that stimulate other endocrine glands. Identify the hormone which specifically acts to trigger hormone secretion by another gland.
(a) Growth hormone
(b) Thyroxine
(c) Adrenocorticotropin hormone
(d) Antidiuretic hormone

107. Which of the following organisms has been commercially utilised as blood cholesterol lowering agent?
(a) *Torulopsis utilis* (b) *Saccharomyces cerevisiae*
(c) *Monascus purpureus* (d) *Pseudomonas putida*

108. Who discovered X–body, but was not able to explain its role ?
(a) Morgan (b) Calvin
(c) Henking (d) Beadle

109. Hyposecretion of anterior pituitary leads to
(a) acromegaly (b) Simmond's disease
(c) Hashimoto's disease (d) Basedow's disease

110. Identify the incorrectly matched pair.

(a) Transformation — Free DNA is incorporated into a recipient cell
(b) Transduction — Donor DNA transfer is mediated by a virus
(c) Conjugation — Transfer of DNA involves cell to cell contact
(d) Sporulation — Formation of cysts

111. The pH of blood in the arteries is 7.4 which is

(a) higher than the pH in veins (b) lower than the pH in veins
(c) equal to the pH in veins (d) varies in different positions

112. Match the following columns (I and II) and select the correct option.

	Column I		Column II
A.	Operator gene	1.	Synthesises a molecule to block a gene adjacent to structural genes
B.	Structural gene	2.	Provides binding site for repressor protein
C.	Promoter gene	3.	DNA sequences that initiate transcription (protein produced a particular gene)
D.	Regulator gene	4.	Provides binding site for activator proteins and RNA polymerase

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	4	1	(b)	2	1	3	4
(c)	3	1	4	2	(d)	3	4	1	2

113. Charging of the *t*RNA is also known as

(a) transcription (b) polymerisation
(c) aminoacylation (d) adenylation

114. Myxoxanthin is found only in

(a) Euglenophyta (b) Cyanophyta
(c) Phaeophyta (d) Pyrrophyta

115. Presence of human chorionic gonadotropin in the urine suggests

(a) Grave's disease (b) goitre
(c) pregnancy (d) diabetes

116. The pattern of cleavage in the human zygote is

(a) determinate (b) holoblastic (c) indeterminate (d) meroblastic

117. Rigor mortis appears first on

(a) neck (b) lower jaw
(c) legs (d) lower abdomen

118. Endarch is

(a) protoxylem lying to the inner side of the metaxylem
(b) protoxylem lying in the middle of metaxylem
(c) protoxylem lying outside the metaxylem
(d) protoxylem lying in the centre of metaxylem

119. Match the regions of the vertebral column correctly with their numbers present in humans.

	Column I		Column II
A.	Cervical	1.	7
B.	Thoracic	2.	1
C.	Lumbar	3.	12
D.	Sacral	4.	5

Codes

	A	B	C	D		A	B	C	D
(a)	1	4	2	3	(b)	4	2	1	3
(c)	1	3	4	2	(d)	4	1	3	2

120. High copy number in a vector is

(a) to provide more replicates of inserted DNA sequences in the cell
(b) easy transformation to the host cells
(c) easy isolation and purification
(d) small size or low molecular weight for enhanced stability

121. Identify the correctly matched pair.

(a) Wet deposition — Acidic water received through rain
(b) Dry deposition — Deposition of wind blown acidic gases
(c) Air pollution — Biomagnification
(d) Both (a) and (b)

122. Which of the following is not exclusively composed of glucose molecules ?

(a) Maltose (b) Glycogen
(c) Lactose (d) Cellulose

123. Quaternary structure of protein is associated with

(a) three-dimensional structure of a peptide chain
(b) orientation of multi-subunits of a protein
(c) amino acid sequence
(d) hydrogen bonding

124. Arrange the following steps in the process of decomposition.

I. Fragmentation II. Catabolism
III. Leaching IV. Humification
V. Mineralisation

Choose the correct option.

(a) I II III IV V
(b) I III II IV V
(c) III I II IV V
(d) III I IV II V

125. The term $(K - N/K)$ is

(a) carrying capacity for any given population
(b) greatest when K is large
(c) zero when population size gets equal to carrying capacity
(d) increased when N approaches K

126. Largest herbarium of the world, National Museum of Natural History is situated in

(a) New York (b) Paris (c) Kew (d) Geneva

127. Which of the following is an example of buffer system in blood?
(a) Oxygen and carbon dioxide
(b) Oxyhaemoglobin and carboxyhaemoglobin
(c) Bicarbonate and carbonic acid
(d) Sodium sulphate and oxalic acid

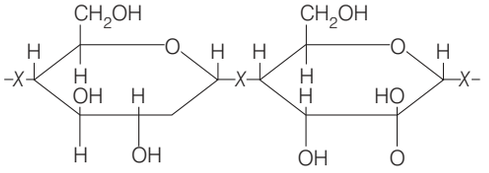
128. Match the tools in Column I with their explanations in Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Shotgun	1.	To make a gene
B.	Antisense genes	2.	To separate fragments of DNA
C.	Gene synthesis	3.	To stop the expression of a gene in a cell
D.	Electrophoresis	4.	To find a particular gene in a whole genome

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	2	1	3
(c)	4	3	1	2	(d)	3	2	1	4

129. Given below is a part of a polysaccharide chain.



Part labelled *X*, represents which type of bond?
(a) —O—, hydroxyl bond (b) —C(=O)—, glycosidic bond
(c) —N(H)—, peptide bond
(d) O=P(O)(O⁻)–O, phosphodiester bond

130. Which of the following is incorrect with respect to plant hormones?
(a) They are synthesised in specific glands and tissues
(b) They are active even in very small concentrations
(c) They are pleiotropic in their effects
(d) Responses to plant hormones are different depending on the tissue or time

131. Which of the following events does not take place during the mitotic prophase?
(a) Disappearance of nuclear envelope
(b) Nucleolar disaggregation
(c) DNA replication
(d) Coiling of chromosome

132. *Zoochlorella* is an example of
(a) epizoic algae (b) endozoic algae
(c) epiphytic algae (d) holophytic algae

133. Match the Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Obligate parasites	1.	Kill the host cells or tissues.
B.	Destructive parasites	2.	Depend upon the host for the supply of the needs of parasite.
C.	Balanced parasites	3.	Normally live as parasites, but become saprophytes under emergent conditions.
D.	Facultative saprophytes	4.	Grow only upon suitable living host tissues.

Codes

	A	B	C	D		A	B	C	D
(a)	1	2	3	4	(b)	4	3	2	1
(c)	1	2	4	3	(d)	4	1	2	3

134. Identify the incorrect statements.
I. Rabbit eats its own faeces.
II. Mouse never drinks water.
III. *Hippopotamus* is called the 'Horse of the river'.
IV. Gibbon is the African dog-like monkey.
(a) I and III (b) II and IV
(c) I and IV (d) II and III

135. Which of the following is not a larval stage of blood fluke ?
(a) Sporocyst (b) Redia (c) Cercaria (d) Miracidium

136. The lymph vessels within the villi help in
(a) absorption of glycerol, fatty acids and monoglycerides
(b) absorption of glucose, galactose and fructose
(c) absorption of fatty acids, glycerols and galactose
(d) absorption of amino acids and monosaccharides

137. Match the Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Negative soil pollution	1.	Reduction in soil productivity due to addition of undesirable substances
B.	Positive soil pollution	2.	Congregating fertile land into barren area
C.	Landscape pollution	3.	Reduction in soil productivity due to erosion and overuse

Codes

	A	B	C		A	B	C
(a)	1	2	3	(b)	1	3	2
(c)	3	1	2	(d)	2	1	3

138. Which variety of silk is produced in Asom ?
(a) Tussar silk (b) Shantung silk
(c) Muga silk (d) Mulberry silk

139. Which of the following is true for F_1-particle of the oxysome ?
(a) It utilises proton energy
(b) It releases proton energy
(c) It has no role related to energy consumption
(d) It is present in the membrane of chloroplast

140. Which statement is true about the climacteric fruits ?
(a) These fruits show increased respiratory activity at the time of ripening
(b) These fruits are developed by artificial method of fertilisation at a certain period of year
(c) These fruits are produced throughout the year in all climates
(d) None of the above

141. The relationship between species *X* and species *Y* is described as commensalism. This indicates that
(a) both species derive mutual benefits like insect pollinators
(b) both species suffer as between *Plasmodium* and its vector
(c) *X* derives benefit while *Y* is harmed like lice living on human body
(d) *X* derives benefit whereas *Y* remains unaffected as between *Remoras* and sharks

142. Find the odd one out in the given list.
I. Diaphragms II. Cervical caps
III. Spremicidal jellies IV. Condoms
V. Nirodh VI. Saheli
VII. Vaults
Choose the correct option.
(a) Only I (b) Only III (c) Only V (d) Only VI

143. The protein present in raw egg white responsible for biotin deficiency is
(a) aleurone (b) riboflavin (c) avidin (d) isoniazid

144. The disadvantage of breeding activities over the technique of GM crops production is
(a) breeding activities can use only those genes, which are present in such species that can be hybridised with them
(b) any gene can be used for transfer
(c) change in genotype can be precisely controlled
(d) produces a desired phenotype

145. A distinctive feature of an insect pollinated plant is that
(a) their pollen grains tend to have rougher surface
(b) they produce sugary fruits
(c) they have nectar, which guides on their flowers
(d) pollen grains are produced in abundance to make up any loss occurring due to wind

146. The reduction of nitrate to nitrite during nitrogen assimilation is carried out by
(a) nitrite reductase and Mg
(b) nitrate reductase and FAD
(c) nitrate reductase and ferredoxin
(d) nitrite reductase and aspartate

147. Which method is used when there are only a very few cells available, such as fertilised egg cells ?
(a) Heat shock (b) Electroporation
(c) Gene gun (d) Microinjection

148. Which of the following cofactors is/are required for the formation of acetyl Co-A?
(a) Mg^{2+} and Co-A
(b) TPP (Thiamine Pyrophosphate)
(c) Lipoic acid
(d) All of the above

149. A sugar is called reducing because of the presence of
(a) a free aldehyde or ketone group
(b) a free oxygen
(c) a free hydroxyl group
(d) None of the above

150. Glissons' capsule and Kupffer cells are present in which of the following parts?
(a) Stomach (b) Gall bladder
(c) Liver (d) Duodenum

151. The transpiration pull overcomes
(a) gravitation pull
(b) resistance of narrow xylem channels and their end walls
(c) resistance offered by water coming out of narrow capillary pores to the soil
(d) All of the above

152. In case of *Gloriosa superba*, autogamy is prevented due to the presence of a barrier in between stamens and style. This is technically known as
(a) dichogamy (b) herkogamy
(c) protogyny (d) heterostyly

153. Which among the following is/are functions of cortex ?
(a) It serves as a protective zone
(b) It helps in absorption and translocation of water
(c) Both (a) and (b)
(d) None of the above

154. A patient had developed characteristic points on teeth called 'Hutchinson's teeth'. Waksman test was used to detect this disease as
(a) chicken pox (b) smallpox
(c) French pox (d) None of these

155. In plants of family Solanaceae, the ovary is
(a) bicarpellary, syncarpous, superior
(b) monocarpellary, syncarpous, superior
(c) tricarpellary, synocarpous, superior
(d) multicarpellary, apocarpous, superior

156. Match the methods of asexual reproduction given in Column I with their explanation given in Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Fragmentation	1.	Simple cell division
B.	Fission	2.	Outgrowth proliferates from thallus
C.	Budding	3.	Arise as modified branches
D.	Gemmae	4.	Thallus breaks down into many fragments

Codes

	A	B	C	D		A	B	C	D
(a)	4	1	2	3	(b)	1	2	3	4
(c)	2	1	4	3	(d)	3	2	1	4

157. *Rhodospirillum* is an example of
(a) cyanobacteria (b) prochlorobacteria
(c) purple or green bacteria (d) mycoplasma

158. Keratinised dead layer of skin is made up of
(a) stratified columnar epithelium
(b) simple columnar epithelium
(c) stratified squamous epithelium
(d) simple cuboidal epithelium

159. Phycomycetes are characterised by
I. Aseptate, coenocytic hyphae.
II. Asexual spores, produced endogenously.
III. Show hetero or isogamous sexual reproduction.
IV. Divided into oomycetes and zygomycetes.
Choose the correct option containing true statements.
(a) I and II (b) I and III
(c) I and IV (d) All are correct

160. Identify the incorrectly matched pair.
(a) Stem rust of wheat — *Puccinia graminis tritici*
(b) Brown rust of wheat — *Puccinia recondita*
(c) Loose smut of wheat — *Puccinia striiformis*
(d) Covered smut of barley— *Ustilago hordei*

161. Which of the following characteristics is never present at any stage of development in primary chordates ?
(a) Post anal tail (b) Pharyngeal slits
(c) Dorsal tubular nerve cord (d) None of these

162. Mark the incorrect statement.
(a) F Redi proposed the theory of special creation
(b) Spallanzani stated that air carried microorganisms
(c) Arrhenius is the father of modern palaeontology
(d) Oparin proposed that life originated on earth by abiogenic molecular evolution

163. Which of the following enzymes promotes the spermatozoan to penetrate the egg during fertilisation ?
(a) Acid phosphatase (b) Hyaluronidase
(c) Amylase (d) Phosphatase

164. Which of the following amino acids has only a single codon ?
(a) Isoleucine (b) Tryptophan
(c) Asparagine (d) Glutamic acid

165. In which of the following can formylmethionine *t*RNA be found ?
(a) Insects (b) Mammals
(c) Bacteria (d) Yeasts

166. Glans of the penis are equivalent to .....in female reproductive system.
(a) vestibule (b) hymen
(c) vagina (d) clitoris

167. The parasites harm the host by
(a) reducing the survival, growth and reproductive ability of the host
(b) reducing its population density
(c) might render the host more vulnerable to predation by making it physically weak
(d) All of the above

168. Name the enzyme involved in ornithine cycle.
(a) Oxidase (b) Dehydrogenase
(c) Amylase (d) Arginase

169. After consumption of a meal containing greasy French fries, which hormones and enzymes would be most active during digestion process?
I. Bile salts, lipase and enterogastrone.
II. Salivary and pancreatic amylase along with disaccharidases.
III. Pepsin, trypsin, chymotrypsin, aminocarboxypeptidases, dipeptidases and CCK.
IV. Salivary amylase, lipase and protein degrading enzymes.
Choose the correct option.
(a) I and II (b) I and III
(c) III and IV (d) II and IV

170. Which among the following causes damages to chloroplast in plants ?
(a) Ozone (b) PAN (c) Phenol (d) Aldehyde

171. Which of the following statements is true for arrhenotoky and thelytoky?
(a) They are forms of sexual reproduction
(b) Arrhenotoky is found in *Peiplaneta* species
(c) In thelytoky, the diploid egg parthenogenetically develops into males
(d) In arrhenotoky, the male develops from unfertilised egg and the female from fertilised egg

172. The causes of biodiversity loss include
(a) habitat fragmentation
(b) overexploitation
(c) intensive agriculture
(d) All of the above

173. Waksman and Woodruff were responsible for the isolation of
(a) actinomycin and streptothricin
(b) penicillin and streptomycin
(c) actinomycin and vancomycin
(d) erythromycin and vancomycin

174. A portion of the endoplasmic reticulum is present in plasmodesmata, it is known as
(a) myeloid body (b) cisternae
(c) vesicles (d) desmotubule

175. Which of the following pigments is not present in the chloroplast?
(a) Anthocyanin (b) Chlorophyll-*b*
(c) Carotenoid (d) Xanthophyll

176. It is not advised to sleep in a closed room with fire burning inside as it could be fatal because
(a) excessive heat
(b) non-availability of oxygen
(c) Hb has higher affinity to CO
(d) combined effect of (b) and (c)

177. Which among the following does not come under the factors affecting primary productivity?
(a) Environmental factor
(b) Nutrients availability
(c) Photosynthetic efficiency of plants
(d) Leaching

178. Which one of the following diseases is due to hypersensitivity?
(a) Hay fever (b) Breakbone fever
(c) Scarlet fever (d) Enteric fever

179. Select the option which is not the true reason for polyembryony in angiosperms.
(a) Cleavage of proembryo
(b) Formation of many embryos due to the presence of more than one embryo sac in the same ovule
(c) Development of many embryos from egg
(d) Formation of many embryos from structures like nucellus and integuments

180. Which one of the following pigments is water soluble?
(a) Chlorophyll-*a* (b) Chlorophyll-*b*
(c) Phycobilin (d) Xanthophyll

